Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Geological Society of America Bulletin

Basin-range tectonics in the Darwin Plateau, southwestern Great Basin, California

EUGENE S. SCHWEIG III

Geological Society of America Bulletin 1989;101;652-662
doi: 10.1130/0016-7606(1989)101<0652:BRTITD>2.3.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Basin-range tectonics in the Darwin Plateau, southwestern Great Basin, California

EUGENE S. SCHWEIG III *Center for Earthquake Research, Memphis State University, Memphis, Tennessee 38152*

ABSTRACT

The Darwin Plateau, in the southwestern Great Basin of California, is underlain by upper Cenozoic basalt flows, pyroclastic rocks, and alluvial deposits. Many of the volcanic units are laterally extensive and can be correlated across the study area, using hand-sample characteristics and magnetostratigraphy. The age of the section, from potassium-argon dating and magnetic polarities, is largely 5.3 to 5.7 m.y., although the oldest basalt flow is 7–8 m.y. old. Structural, radiometric, and sedimentologic data all suggest that basin-range crustal extension was underway locally by 7–8 Ma, and the earliest high-angle normal faulting predates 5.8 Ma. This chronology is consistent with westward migration of tectonism in the southwestern Great Basin. Additionally, structural data suggest two extension directions, the current west-northwest–east-southeast direction and an earlier west-southwest–east-northeast one. The change in extension direction occurred after 5.7 Ma. Local and regional data also indicate that the maximum compressive and intermediate stresses in the latest stress regime have been approximately equal in magnitude. Paleomagnetic data do not indicate significant rotation of the Darwin Plateau between the many large strike-slip faults in the area.

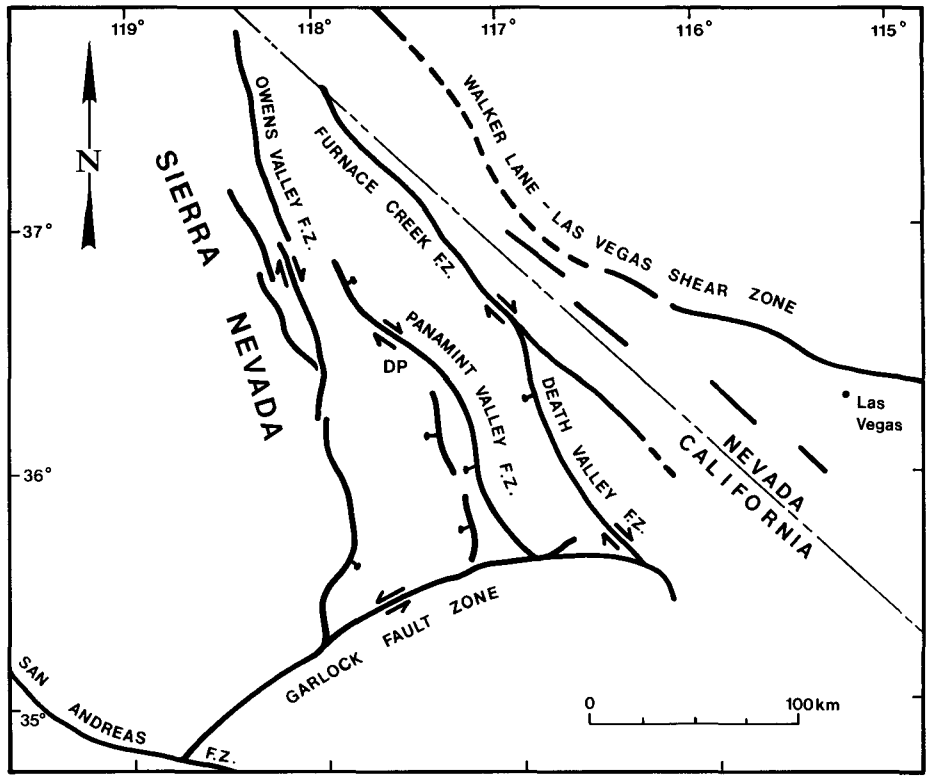


Figure 1. Major faults of the southwestern Great Basin. DP, Darwin Plateau.

INTRODUCTION

The southwestern Great Basin is a subprovince of the Basin and Range, lying north of the Garlock fault, east of the Sierra Nevada, and west of the Walker Lane belt of sigmoidal bending (Fig. 1). The southwestern Great Basin differs from the rest of the province in its topography, structure, seismicity, and current stress field (for example, Wright, 1976).

This study focuses on the Darwin Plateau in the southwestern Great Basin of east-central California (Fig. 2). It is one of many elevated areas of subdued topography in the ranges of the region. The Darwin Plateau has an average elevation of about 1,400 m and stands 900 m above the Panamint Valley floor and 400 m below the crest of the Argus Range.

In this study, the volcanic and alluvial stratigraphy and the structural geology of the Rainbow Canyon area on the east side of the Darwin Plateau (Fig. 3) are examined, with the goal of determining the Neogene tectonic history of the area. Although the study area is small, its central location within the southwestern Great Basin and its excellent exposure of structure and the thick Neogene sedimentary and volcanic section provide data pertinent to several important geologic problems. These include the age of local and regional inception of basin-range tectonism, the direction of local extension, and whether major block rotation has occurred during regional strike-slip faulting.

The inception of basin-range tectonism can be examined in terms of three criteria. One is the formation of the modern fault-controlled basins and associated gross topography (Stewart, 1978; Zoback and others, 1981). The inception of high-angle normal faulting may also be a good indicator (Eaton, 1982). Third, the beginning of basin-range crustal extension may be marked by a change from calc-alkaline intermediate volcanism to basaltic or bimodal basaltic/rhyolitic volcanism (for example, Christiansen and Lipman, 1972). Normal faulting and volcanism in the Darwin Plateau area are clearly too young to

Geological Society of America Bulletin, v. 101, p. 652–662, 11 figs., 2 tables, May 1989.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

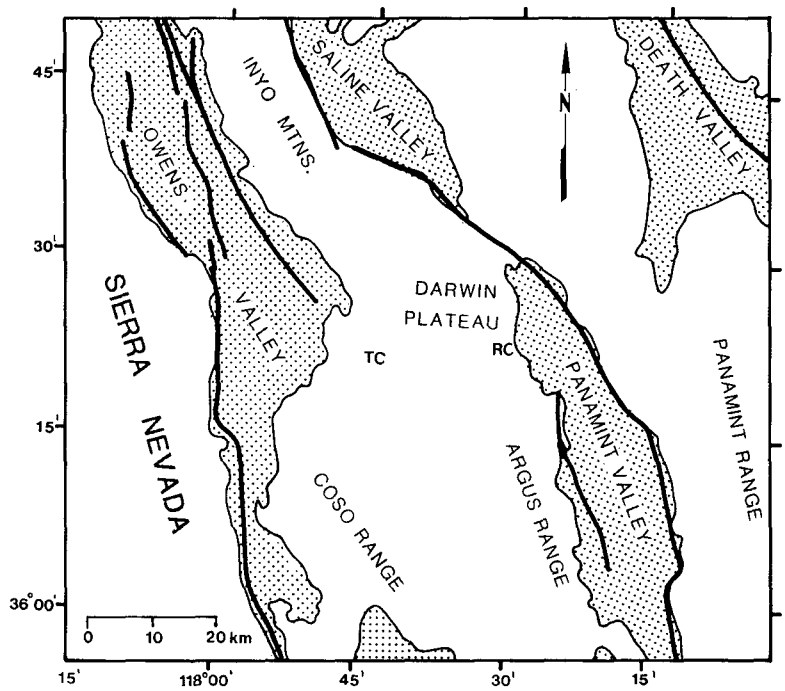


Figure 2. Principal physiographic elements of the region surrounding the Darwin Plateau. Stippling indicates areas of Quaternary fill. TC, Talc City Hills; RC, Rainbow Canyon.

display the complications of pre–basin-range extension seen elsewhere in the Great Basin, making application of the three criteria here simpler than elsewhere.

The Darwin Plateau has been included in several earlier geologic studies, but none of this work has focused on late Cenozoic history (for example, Hopper, 1947; Hall and MacKevett, 1962; Hall, 1971). Larsen (1979) has obtained whole-rock K-Ar dates and chemical analyses on basaltic rocks in a large area that includes the Darwin Plateau. Recent deformation in the Saline Valley region, immediately northwest of the Darwin Plateau, has been investigated by Zellmer (1980).

STRATIGRAPHY

The Neogene volcanic and sedimentary rocks that underlie the Darwin Plateau are well exposed in the canyons that drain into Panamint Valley. The best exposures appear in the walls of Rainbow Canyon (Fig. 3). A geologic map of the Rainbow Canyon area is presented in Figure 4. Seven stratigraphic sections were measured and described in and adjacent to Rainbow Canyon. These sections and a composite stratigraphic column are shown in Figure 5.

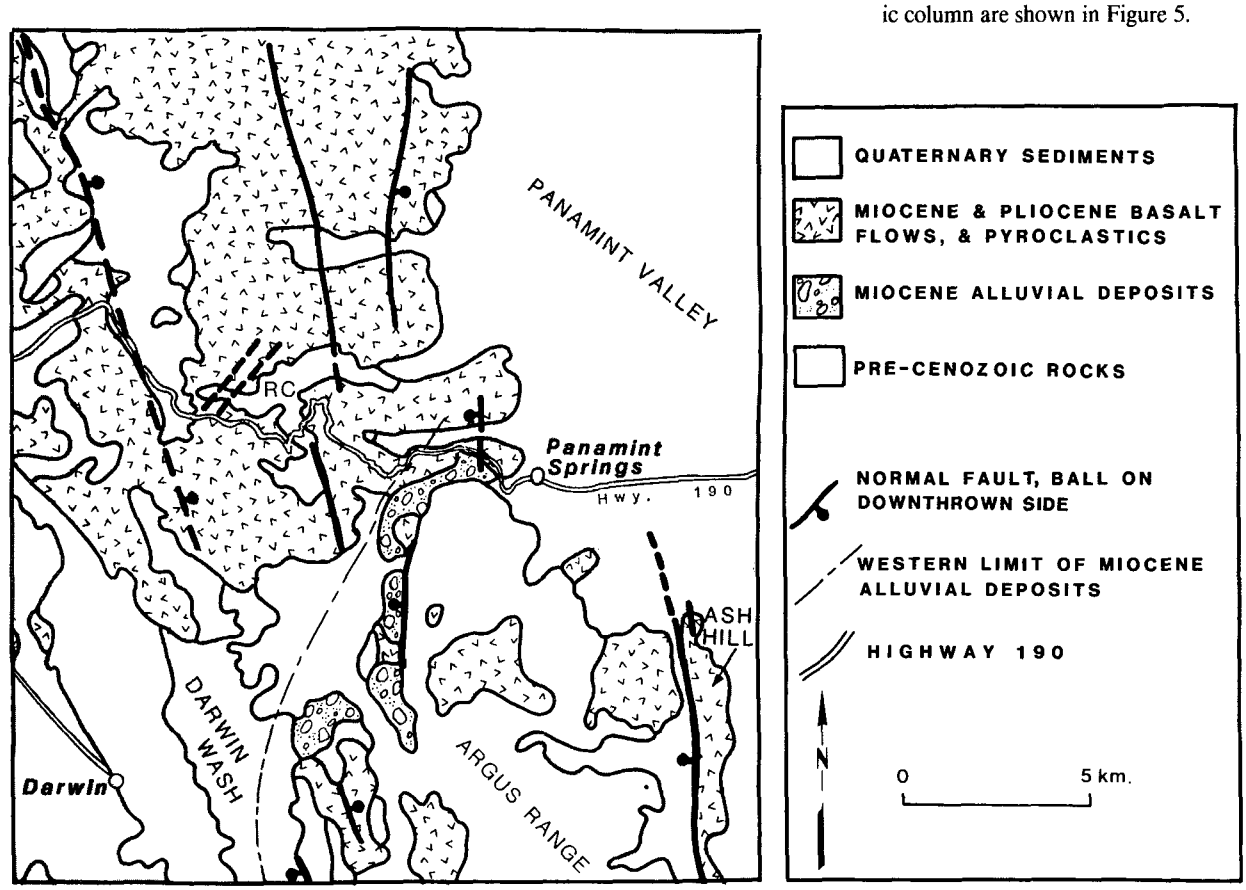


Figure 3. Generalized geologic map of the eastern Darwin Plateau (Hall and MacKevett, 1962; Hall, 1971; Schweig, 1982). RC, Rainbow Canyon.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

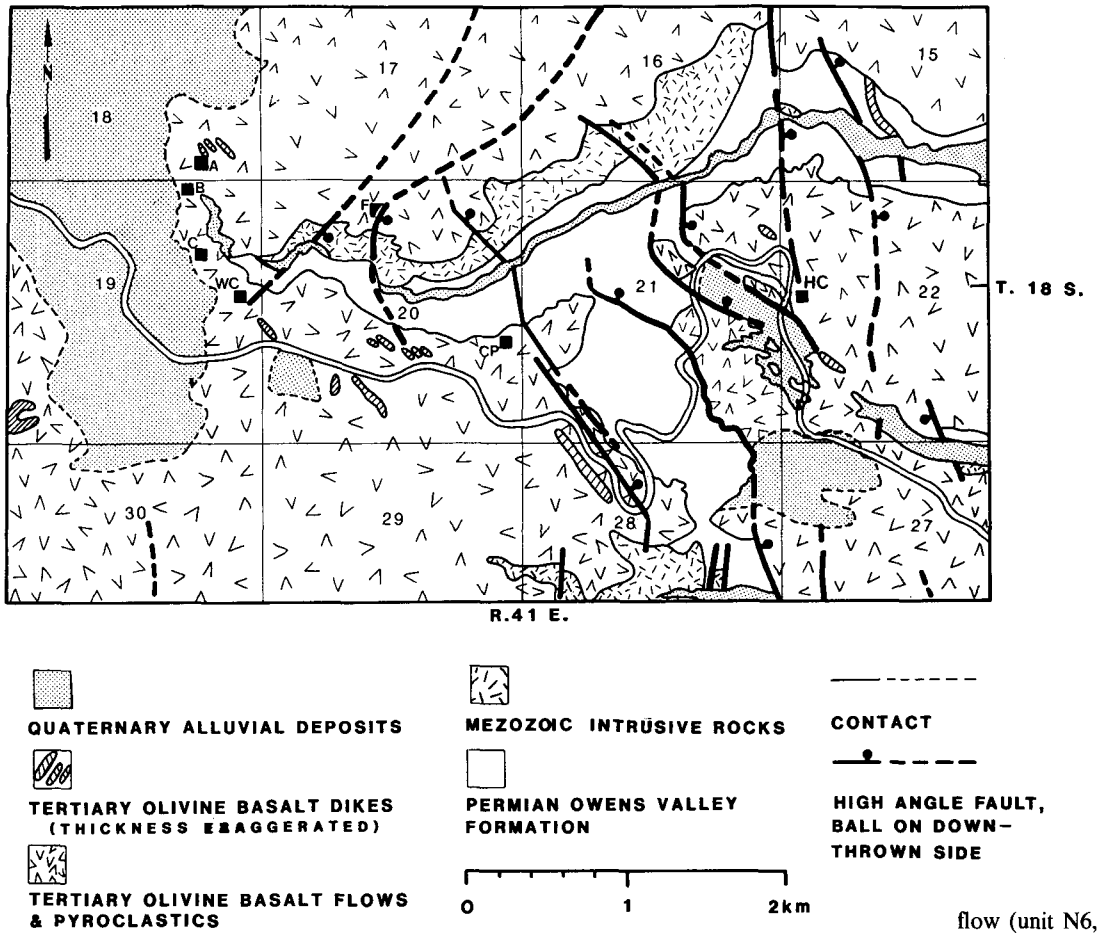


Figure 4. Generalized geologic map of the Rainbow Canyon area. Letters with squares refer to locations of measured sections (Fig. 5). Thicknesses of dikes exaggerated.

In general, Neogene rocks rest unconformably on a fairly flat erosion surface developed across turbidites of the Permian Owens Valley Formation and granitic rocks of probable Jurassic age (Hall and MacKevett, 1962; Hall, 1971). Along the eastern margin of the Darwin Plateau and the Argus Range, the lowermost Cenozoic units are a welded rhyolite tuff and fanglomerate (fanglomerate no. 3 of Hall, 1971). The tuff forms the basal Tertiary unit in many outcrops and presumably represents remnants of a once widespread unit. It is at least 45 m thick.

Unconformably overlying the rhyolite tuff is the highly indurated fanglomerate. It is at least 16 m thick along California State Highway 190, about 1.2 km west of Panamint Springs. Most of the clasts are probably of local derivation, but clasts of schistose and gneissic rocks are also common. About 3% of the total clast population consists of an extremely well-sorted and pure quartzite, closely resembling the Eureka Quartzite that crops out in the Panamint Range to the east and in the Talc City Hills to the west (Fig. 2). The other metamorphic clasts almost certainly were derived from more easterly sources in the Precambrian rocks of the west-central Panamint Range; many of the plutonic clasts are of types not known to be exposed in the Argus Range. Both the old fanglomerate and the rhyolite tuff are tilted 20° to 40° to the east and are found only east of the dashed line in Figure 3. Thus, when these rocks were deposited, Panamint Valley had not yet formed, but the Panamint Range had been uplifted and was eroding. The tuff and the old fanglomerate probably extended farther to the west prior to eastward tilting of the Argus Range and its accompanying erosion.

The remainder of the Cenozoic section in the Rainbow Canyon area is composed of olivine basalt flows, basaltic lapilli tuffs and tuff breccias, and alluvial deposits (Fig. 5). Basaltic flows and pyroclastic rocks either are exposed at the surface or lie beneath a thin veneer of Quaternary alluvium throughout the Rainbow Canyon area except in canyon bottoms and in section 21 (Fig. 4) where basalt has been removed by erosion. Individual basaltic sequences are as much as 160 m thick, and most are composed of more than 90% basaltic flow rocks.

Many of the flows, particularly in the lower part of the section, are areally extensive. One flow (unit N6, Fig. 5) can be traced with certainty for 2.5 km and probably for at least 7 km. Flows range in thickness from less than 1 m to more than 60 m. The thickness of a given flow varies little over distances of a kilometer, and the terminations of flows, where exposed, are abrupt.

Although there are many eruptive centers in the vicinity of Rainbow Canyon, as identified by clusters of cinder cones and basaltic dikes, it is difficult to correlate them with individual flows. Indeed, the sources for the thick flows in the lower two-thirds of the section may be outside the area shown in Figure 4.

Layers of lapilli tuff and tuff breccia as much as 9 m thick commonly are present between basalt flows. The thicker layers comprise as many as seven separate eruptive units, commonly with evidence of erosion or soil development between units. In places where the pyroclastic rocks can be traced to their source areas, the clast sizes and thicknesses of the deposits increase greatly and rather abruptly near the vents. For most of the pyroclastic layers, however, the thickness is constant throughout the map area shown in Figure 4, suggesting distant sources.

Hand-sample and thin-section examination of the basalt flows from the Rainbow Canyon area

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

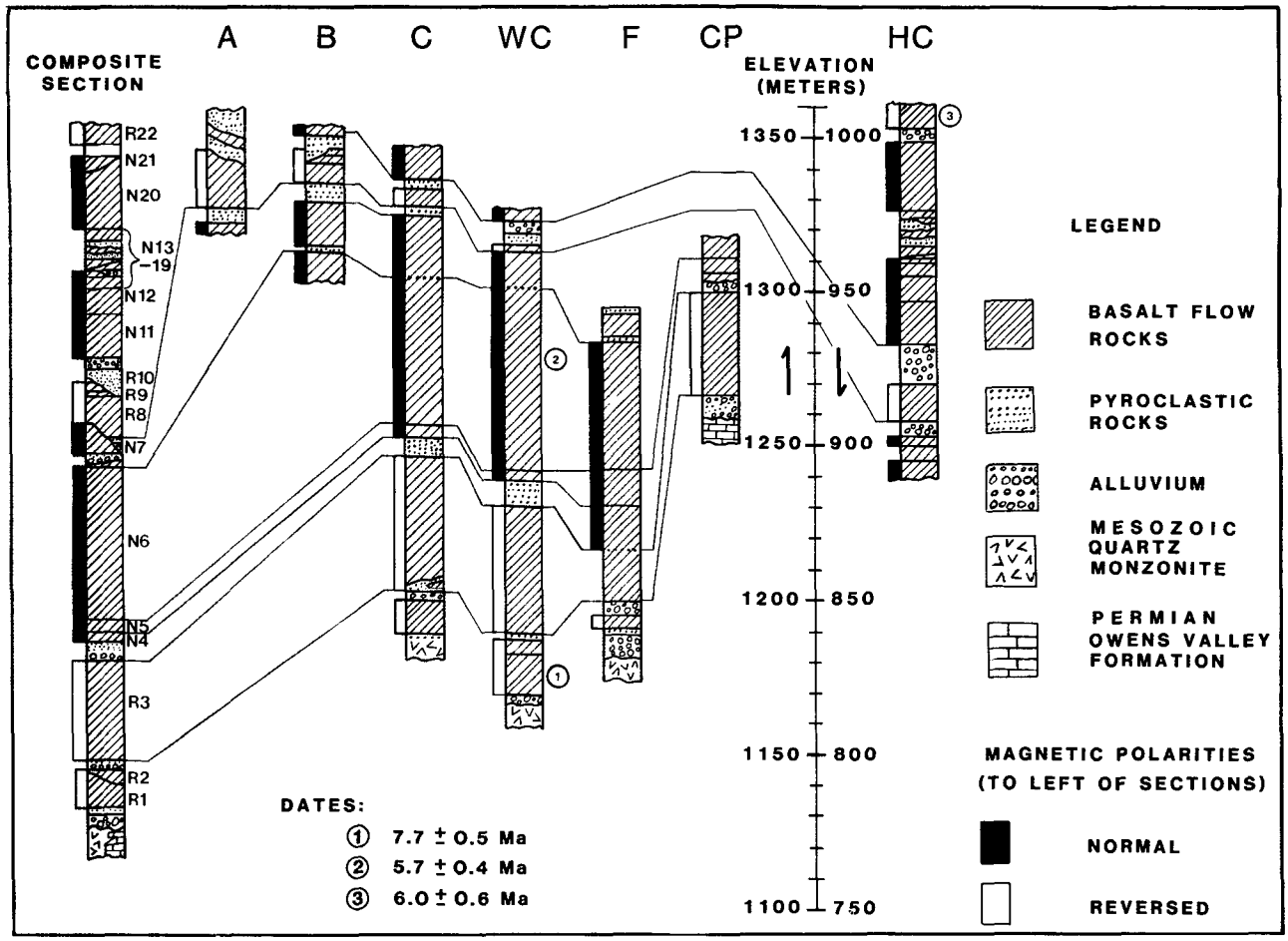


Figure 5. Stratigraphic columns for the Rainbow Canyon area. Locations of sections shown in Figure 4. Composite section shown on left with unit names.

indicates that all are olivine basalt. Larsen (1979) chemically analyzed four samples from localities in or near Rainbow Canyon. Two of the samples were identified as calc-alkaline high-alumina basalt and two as alkali-olivine basalt according to the classification scheme of Irvine and Baragar (1971). It was not possible to correlate Larsen's samples with the units shown in Figure 4.

Magnetic Polarities and K/Ar Ages of Basalts

During this study, the magnetic polarity of each basalt flow was determined in the field, using a hand-held fluxgate magnetometer according to the method of Doell and Cox (1967). The data were used to correlate units across the Rainbow Canyon area. The results are shown in Figure 5. Units that are normally polarized are prefixed by N; those that are reversely polarized, by R.

At least five magnetic-polarity intervals are represented in the basaltic section. Because the time intervals between successive flows are not known, it is not known whether additional reversals of the magnetic field occurred during gaps in the rock record.

Five potassium-argon ages are available for rocks exposed within the mapped area shown in Figure 4 (Table 1). Three of the samples were collected for this study from Rainbow Canyon and the small canyon to the south (Schweig, 1982). The locations of the samples and their positions within the stratigraphic column are shown in Figure 5. These age determinations date unit R1 at 7.7 ± 0.5 Ma, N6 at 5.7 ± 0.4 Ma, and R22 at 6.0 ± 0.6 Ma (the error limit is one standard deviation about the mean). Including the error limits, the ages are stratigraphically consistent.

If it is assumed that the ages of units N6 and R22 lie within their analytical error limits, estimates of the ages of the rocks can be made (Fig. 6). The age range for reversely magnetized unit R22, the youngest stratigraphically, puts it in reversed chrons 3Br, 3Ar, 3r, or subchron 3A-1, according to the magnetic-polarity time scale of Harland and others (1982). The range for unit N6, which is normally polarized, puts it in normal chron 3A. Because unit N6 is stratigraphically below unit R22, both units must have been formed during the overlap period of their possible age ranges (5.4 to 6.1 Ma). Units N6 and R22 are separated, however, by a reversely polarized section (units R8, R9, and R10) and a normally polarized section (units N11 through N21). As shown in Figure 6, there is only one way to correlate this polarity pattern with the polarity time scale. Thus, the age of unit N6 (and therefore unit N7) is 5.8 to 6.1 m.y., and the age of unit R22 is about 5.4 m.y. Intervening units R8 through R10 would be 5.7 to 5.8 m.y., and units N11 through N21 would be 5.4 to 5.7 m.y. These age assignments are consistent with field relations; evidence of erosion or alluvial deposition, or their absence, suggests that the flows within each polarity group are very close in age and that there were significant

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

TABLE 1. POTASSIUM-ARGON AGE DETERMINATIONS FOR RAINBOW CANYON

No.	Sample name	Rock type	Location	Calculated age (m.y.)	Reference
1	GRC-72	Olivine basalt	Unit R1, Rainbow Canyon	7.7 ± 0.5	This study
2	HRC-82	Olivine basalt	Unit N6, Rainbow Canyon	5.7 ± 0.4	This study
3	QHC-142	Olivine basalt	Unit R22, Rainbow Canyon	6.0 ± 0.6	This study
4	DV-4	High-Al basalt	Rainbow Canyon area	5.3 ± 0.2	Larsen (1979)
5	DV-21B	Alkali basalt	Rainbow Canyon	4.3 ± 0.2	Larsen (1979)

Note: all dates on whole-rock material. Errors are ±1σ.
Samoles for this study analyzed by Geochron Laboratories, Cambridge, Massachusetts. Details given in Schweig (1982).
Dates corrected to constants of Steiger and Jäger (1977): $\lambda_\epsilon = 0.581 \times 10^{-10}yr^{-1}$, $\lambda_\beta = 4.692 \times 10^{-10}yr^{-1}$, $^{40}K/K = 1.167 \times 10^{-4}$ mol/mol.

hiatuses in eruptive activity between groups. Even if 2σ error limits were used, the age of the N6–R22 package would be constrained to 4.8 to 6.5 m.y. and would not significantly affect the results of the study.

The magnetic polarities of the lowermost five units (R1–N5) do not help to constrain their ages. The ages of these five units are constrained only to fall between 5.8 and 8.2 Ma. No erosion or deposition, however, is found between units N5 and N6. These two units are thus considered to be the same age.

The other two age determinations from the Rainbow Canyon area are by Larsen (1979). The stratigraphic positions of age determination 5 within the section are not certain, and age determination 4 is on a flow isolated from the stratigraphic sectins shown in Figure 5. Their ages, however, are younger than those of any of the basalt flows in the measured sections of this study.

Basaltic Dikes

Of the basaltic dikes shown in Figure 4, the thirteen that are accessible show average surface trends of N30°W. No absolute ages are yet available for the dike rocks. Field relations suggest that the basalt flow dated at 5.3 ± 0.2 Ma (age determination 4, Table 1) is probably of about the same age as the long dike in the northwest quarter of section 28 (Fig. 4). All of the dikes that were examined are younger than unit R10 (about 5.7 to 5.8 m.y. old), and they cut all of the units exposed in the areas where they occur. Dikes in sections 19 and 20 (Fig. 4) apparently fed the same cinder cone from which unit N11 (5.4 to 5.7 m.y. old) emanated. The easternmost two dikes are younger than unit R22 (about 5.4 m.y. old).

These age relations suggest that the dikes were formed about 5.3 to 5.7 Ma, and perhaps entirely after 5.4 Ma. This probably was the period of eruptive activity within the map area shown in Figure 4. No crosscutting relations were observed between basaltic dikes and faults, nor between any two dikes.

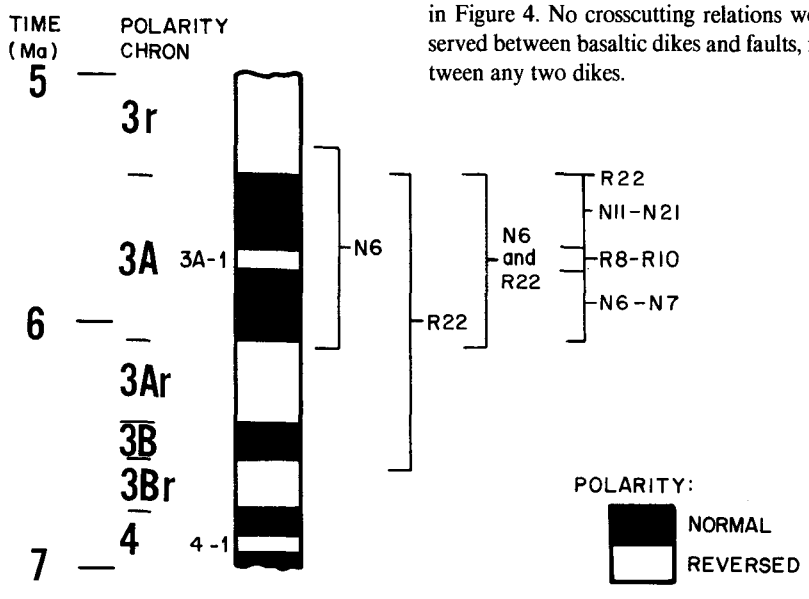


Figure 6. Derivation for ages of units N6 through R22. Magnetic polarity time scale after Harland and others (1982).

STRUCTURE

Faults in the Rainbow Canyon area are widely distributed and display a fairly simple pattern, particularly when compared to those east of Panamint Valley. The faults strike north-northwest to east-northeast, and most of them show normal displacement.

Many small faults, those with as much as several meters of throw, are visible in the alluvial layers in the Rainbow Canyon area. The average strike of the small faults is about N20°E. This is typical of the faulting pattern throughout the southwestern Great Basin (for example, Wright, 1976). Most of these faults dip 65° to 75°, in general toward the east. Seventy-seven of these features show striations or other evidence of the direction of slip. Small faults that strike north-northwest commonly have very steep dips (>75°). Their surfaces are marked by striations that plunge at angles less than 30°, implying that their latest slip was mainly horizontal.

Large faults, with many meters of net slip, strike north-northwest and northeast. The sense of dip-slip on all of these faults is down to the east. Evidence of some lateral slip along at least two of these faults was observed on the north wall of Rainbow Canyon, where unit thicknesses do not match. One of these faults strikes northeast, the other north-northwest.

The greatest throw measured on Tertiary units along a fault is 161 m down to the east on the more westerly of two faults in Rainbow Canyon shown in the northwest quarter of section 21 (Fig. 4). The total throw on the base of the Tertiary section on this pair of faults is 277 m.

The more easterly of these two faults separates the base of the Tertiary section 116 m vertically. The entire basaltic section is cut by the fault, the offset of which is thus at least in part younger than 5.4 m.y. Overlying the basaltic section is a variegated orange and green lapilli tuff that is offset only ~20 cm. Red cinders deposited above the variegated lapilli tuff have not been offset by the fault. Unfortunately, none of these pyroclastic rocks is suitable for isotopic dating. Movement on this fault, however, must have ceased before the cessation of the latest period of local volcanism. Enough time passed between faulting and the development of the overlying cinder cone to allow for removal of the fault scarp through erosion and the formation of a talus apron.

Figure 7 shows a fault in the center of section 20 (Fig. 4) as it appears in the north wall of Rainbow Canyon. One branch of this fault terminates below unit N5, which flowed over the fault scarp. Field evidence suggests that unit N5 was formed at about the same time as unit N6

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

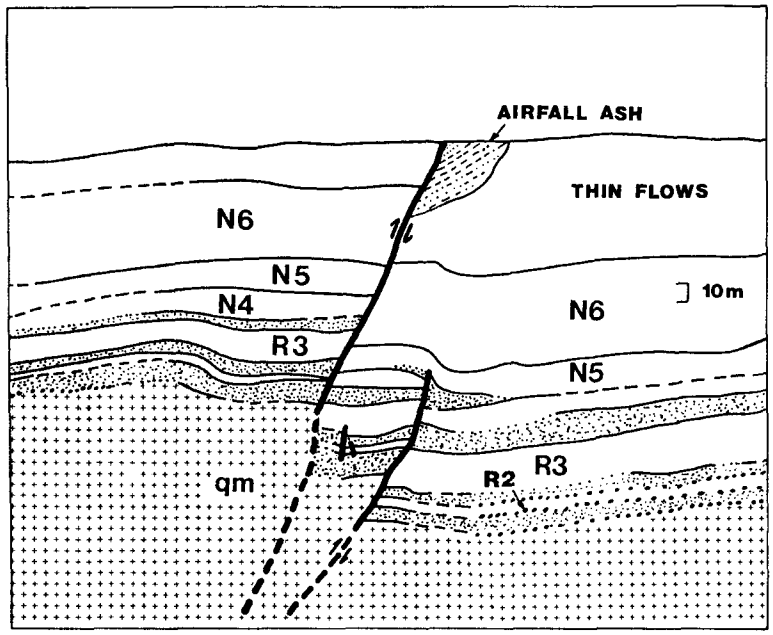


Figure 7. Sketch of relations in north wall of Rainbow Canyon as seen from south rim. Faults have normal slip but appear reversed owing to angle of view. Stippled pattern represents alluvium, and pattern with "qm" represents quartz monzonite.

(5.8 to 6.1 Ma). Normal faulting in the Rainbow Canyon area thus must have begun by this time. In addition, the throw is about 35 m for all the units cut by the throughgoing branch of this fault; hence, no progressive offset can be demonstrated. Accessibility was insufficient to permit the measurement of progressive offsets on the east branch of the fault, but the throw of the lowest units is about 20 m. Therefore, of the 55 m of total vertical offset on this fault, 20 m took place prior to 5.8–6.1 Ma, and 35 m occurred later than 5.4 Ma. This fault probably strikes north-northeast, as shown by the slight topographic lineament extending away from the north side of the canyon, but its continuation on the south side of the canyon is unclear.

There are many large faults and suspected faults cutting the Cenozoic section that are outside the area shown in Figure 4 but within or adjacent to the Darwin Plateau. These faults, with lengths of several to tens of kilometers, consistently strike north-northeast to north-northwest. Almost all of them show apparent normal offsets of tens of meters. One of these, the northwest extension of the Panamint Valley fault zone across Hunter Mountain that connects with faults on the west side of Saline Valley, appears to be very young in geomorphic terms. As suggested by Zellmer (1980) and Burchfiel and others (1987), this fault seems to be largely strike slip, transforming oblique, down-to-the-west slip on the Panamint Valley fault zone into down-to-the-east normal faulting in Saline Valley.

TECTONIC HISTORY

A synthesis of the preceding data suggests the following sequence of events for the Rainbow Canyon area (Fig. 8):

Prior to 7.7 Ma:

1. Erosional development of a moderately smooth ground surface and deposition of locally derived patches of monolithologic alluvium.

2. Deposition of rhyolite tuff and older, Panamint Range–derived alluvium, at least in the eastern part of the study area.

7.7 to 5.8 Ma:

3. Eruption of units R1 through N5 (7.7 to 6.1 Ma) and N6 through N7 (6.1 to 5.8 Ma) as thick, laterally extensive flows and pyroclastic units, all with unknown sources but probably derived from the southwest. Intervals of erosion between successive flows (except, perhaps, flows N5 and N6). At least three intervals of alluvial deposition (above flows R2, R3, and N6). High-angle normal faulting beginning by the time flow N5 was erupted.

5.8 to 5.7 Ma:

4. Erosion, completely removing unit N7 at some localities. Deposition of some patches of alluvium.

5. Eruption of units R8, R9, and R10, probably from the vicinity of sections 17 and 18 as shown in Figure 4.

6. Eruption of pyroclastic materials from the area of sections 19 and 20, blanketing the entire area shown in Figure 4. Possible beginning of injection of north-northwest–trending basaltic dikes.

7. Local deposition of alluvial gravels.

5.7 to 5.4 Ma:

8. Eruption of flows N11 through N14, from unknown source areas.

9. Eruption of thin flows N15 through N21 from the vicinity of the northwest quarter of section 21 as shown in Figure 4.

10. Widespread deposition of alluvial gravels and sands.

About 5.4 Ma:

11. Eruption of unit R22.

Movement on at least one fault had ceased and its fault scarp had been completely eroded before the end of local volcanism.

Inception of Basin-Range Extension

As mentioned previously, the inception of basin-range crustal extension on the Darwin Plateau can be defined by the formation of the modern fault-controlled basins and associated gross topography, by the inception of high-angle normal fauting, and by the beginning of basaltic volcanism.

The first criterion can be applied to the older, Panamint Range–derived fanglomerate exposed east of the dash-dot line shown in Figure 3. These rocks were derived, in part, from eastward sources in the Panamint Range, indicating that uplift had begun east of the Darwin Plateau area. This fanglomerate mostly predates the oldest dated basalt in the Panamint Range (5.13 ± 0.35 Ma; Hall, 1971), as it contains no basaltic clasts except in the top 1 m. Angular discordance between these alluvial deposits and overlying basalt (unit R3) suggests that the older alluvium also was deposited before 7.7 to 6.1 Ma. The end of deposition of this alluvium is marked by a change from debris flows to stream-laid deposits that contain clasts of local derivation, perhaps reflecting initial eastward tilting of the Argus Range. This occurred before 7.7 to 6.1 Ma, but probably only shortly before.

High-angle normal faulting, the second indicator of basin-range extension, dates from before 5.8 to 6.1 Ma on the Darwin Plateau. This is evidenced by the fault shown in Figure 7. For the third indicator of extension, the oldest Neogene basalt in the Rainbow Canyon area, and apparently the oldest yet dated west of Death Valley, is the basal 7.7 ± 0.5 Ma flow in Rainbow Canyon.

The data from all three criteria are consistent with the inception of basin-range faulting and the formation of the Argus Range occurring by

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

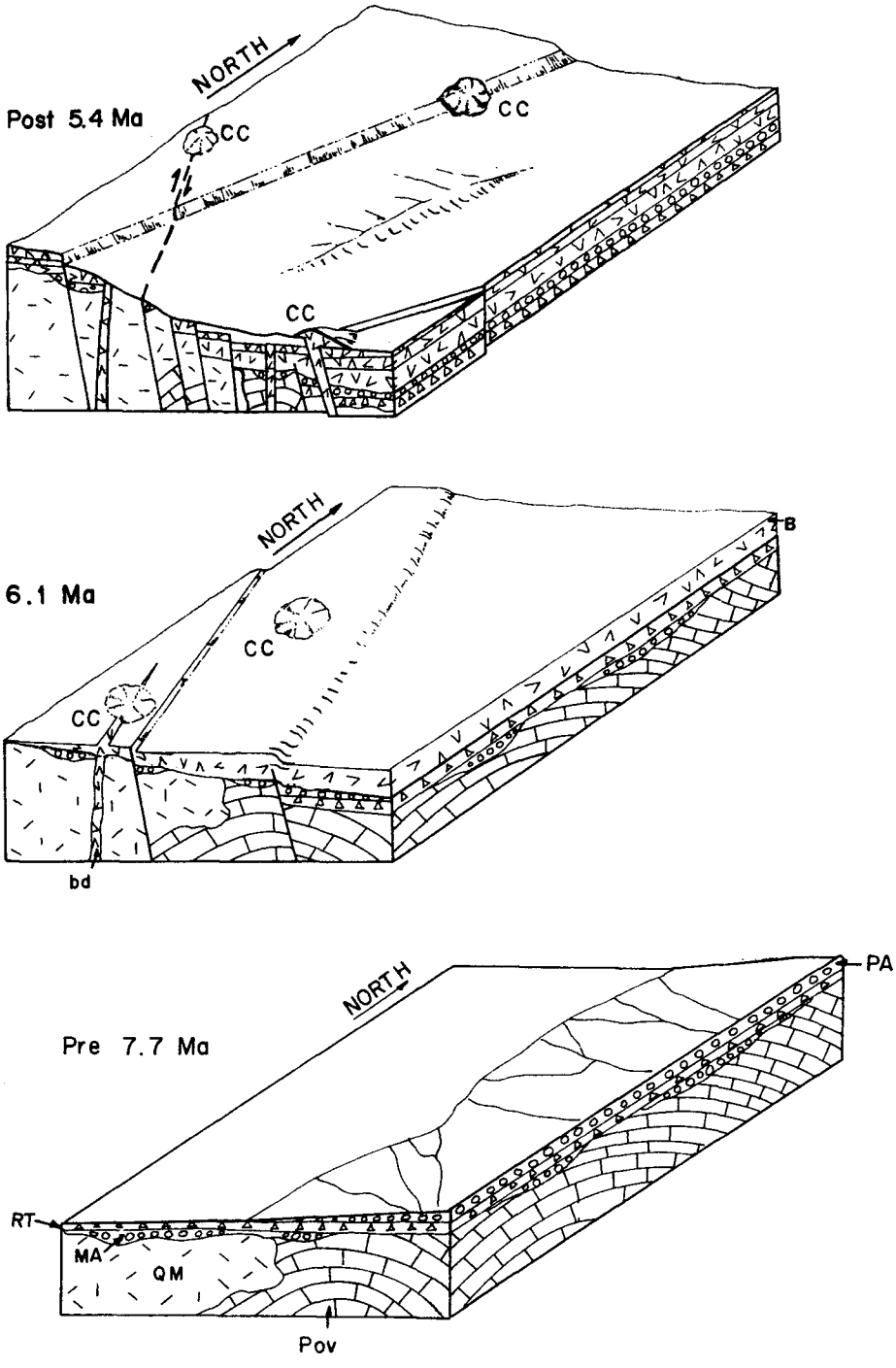


Figure 8. Schematic development of the Rainbow Canyon area. QM, quartz monzonite; Pov, Owens Valley Formation; MA, monolithologic alluvium; RT, rhyolite tuff; PA, alluvium from Panamint Range; B, basalt flows; bd, basalt dike; cc, basaltic cinder cone.

about 7–8 Ma. Some lag time is possible between initial breakup and the oldest exposed related sedimentary deposits. Probably, there are older faults not easily dated, and possibly, there are older, unexposed basalt flows. It should be noted that this age for the inception of basin-range tectonism is older than that suggested by the Massachusetts Institute of Technology 1985 Field Geophysics Course and Biehler (1987) for the opening of Panamint Valley. They have stated that if the 4 Ma basalts on the Darwin Plateau and in the Towne Pass area were once continuous, then the two sides of Panamint Valley separated along a low-angle normal fault after 4 Ma because no basalt is indicated beneath shallow Panamint Valley fill. If their interpretation is correct, then the actual opening of Panamint Valley lagged behind initial crustal extension by several million years.

The age of inception of basin-range tectonics in the Darwin Plateau area agrees well with the suggestion of Wright and others (1984) and Schweig (1986) that there has been a westward migration of tectonism in the southwestern Great Basin. East of the Panamint Range, the interval of most intense crustal extension shows a westward decrease in age ranging from older than 14 m.y. in the Kingston Range to 8 m.y. and younger in the Black and Greenwater Ranges (Wright and others, 1984). West of the Darwin Plateau, in the southern Inyo Mountains, volcanism and faulting appear to have begun around 6 Ma (Bacon and others, 1982). About 4 Ma, tectonism spread north and south into the Coso and Saline Ranges.

DIRECTION OF CRUSTAL EXTENSION

The structural data discussed earlier can be used to make some inferences as to the direction of crustal extension on the Darwin Plateau. Recently, Michael (1984) presented a technique to invert directional data to estimate the stress field that caused the faulting. Using the technique, one assumes that the direction of the tangential traction on a fault plane parallels the slip direction determined from the slickenside data, that all slip events are independent and represent the same stress tensor, and that the magnitude of the tangential traction is a constant on the different fault planes. A requirement of this method is the specification of a unique direction of slip on each fault.

This technique was used on 59 small faults in Neogene rocks in or near Rainbow Canyon for which unique slip directions could be determined or inferred. Although it appears likely from the data (Fig. 9A) that some of the faults violate the second assumption (that is, more than one stress tensor is involved or there has been some block interaction), analysis of the second data set found a least compressive stress, S_3, oriented about S82°W-N82°E and a maximum compressive stress, S_1, to be nearly vertical (Fig. 9B), consistent with a normal-fault regime.

Figures 9C and 9D show the results of an analysis of a subset of 15 small faults in the south wall at the west end of Rainbow Canyon. All of these faults are exposed within the sedimentary units below basalts R1 and R2. The

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

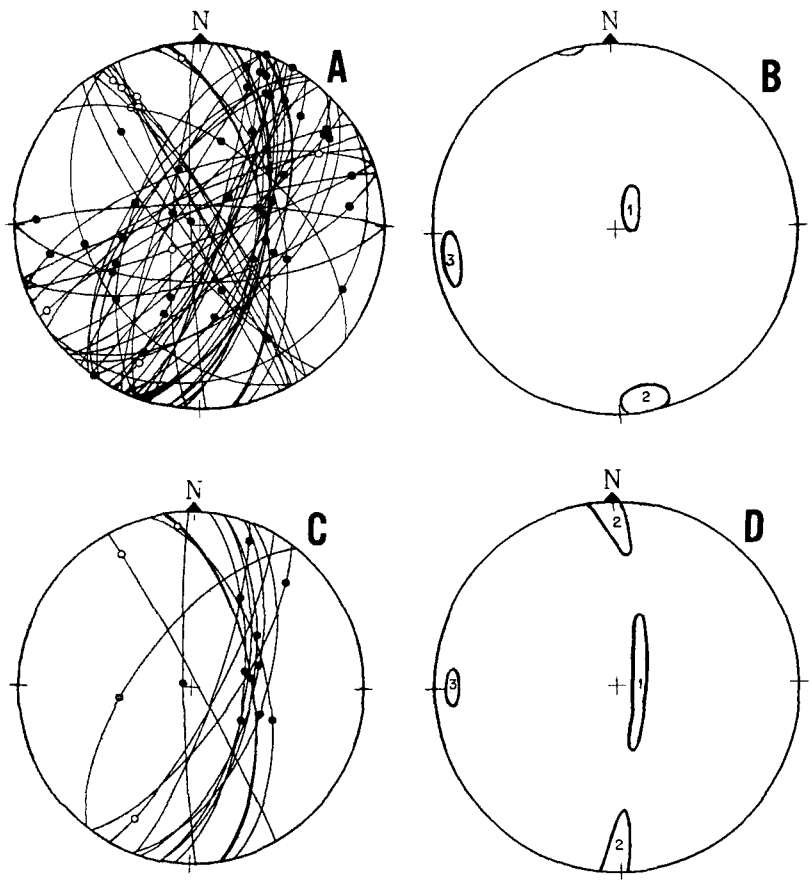


Figure 9. Lower-hemisphere stereonet plots. (A) Fault planes and slip directions from slickensides in the Rainbow Canyon area. (B) Three principal stress directions from data in A and their 95% confidence areas. (C) Fault planes and slip directions from the southern wall at the west end of Rainbow Canyon. (D) Three principal stress directions from data in C. Open circles represent thrust faulting in A and C. In B and D, 1 is the maximum compressive, 2 the intermediate, and 3 the maximum extensional stress.

orientations in this small set of faults are consistent with formation by a single stress tensor. Only faults for which the direction of slip was completely unambiguous were included. For this subset, S_3 was determined to be about N87°W-S87°E. The direction for S_1 is similar to that in the previous analysis. The north-south elongation in the 95% confidence areas of S_1 and S_2 is one indicator that their magnitudes are very close. This tensor, with S_3 oriented slightly north of east-west with S_1 and S_2 very close in magnitude, is consistent with regional faulting patterns showing north- to north-northeast–striking normal faults both locally and regionally, as well as with the major north-northwest–striking faults of the region with evidence of a right-lateral component of slip (for example, Smith, 1975; Roquemore, 1980; Zoback and Zoback, 1980, 1989; Lubetkin and Clark, 1985; Burchfiel and others, 1987). It is therefore thought to represent the most recent stress field. Other data subsets from small areas in Rainbow Canyon also showed this north-south elongation and contained steeply dipping, north-northwest–striking faults with shallowly plunging striations.

The 5.3 to 5.7 Ma dikes are suggestive of another extension direction. Surface trends of 12 out of 13 dikes are northwest to north-northwest, with an average of about N30°W. Zoback and Zoback (1980) have pointed out that studies comparing dike trends with other indicators of extension direction indicate that dikes are perpendicular to the S_3 direction and tend to crosscut pre-existing faults. The dikes on the Darwin Plateau are thus indicative of local west-southwest–east-northeast extension. Also consistent with a period of west-southwest–east-northeast extension are the north-northwest–striking large normal faults and the fraction of the normal faults in Figure 9A that strikes north-northwest and is inconsistent with the overall stress tensor suggested by those data. The total data set may therefore represent a combination of faults formed under both stress fields.

If valid, these data suggest that a period of west-southwest–east-northeast extension with associated emplacement of north-northwest–striking dikes and development of north-northwest–striking normal faults began by step 3 and continued at least through step 10 of the history given above. West-northwest–east-southeast extension with a lateral component of slip on north-northwest–striking faults and normal slip on north- to northeast-striking faults began after step 10 and continues to the present.

Zoback and Zoback (1989) have considered analysis of small-scale faults to result in A quality data (accurate to ±15°), whereas the large-scale faults provide C quality data (accurate to ±30°). If the accuracies of the data collected are those indicated by Zoback and Zoback (1989), the extension directions suggested here are N60°E-S60°W ± 15° from the dikes, followed by N87°W-S87°E ± 15° from the fault subset analysis. These two directions are significantly different. A switch from west-southwest–east-northeast to west-northwest–east-southeast extension is seen in many places in the Basin and Range province (Zoback and others, 1981). In central Nevada, Zoback and Thompson (1978) were able to constrain this clockwise change in extension directions to between 14 and 6 Ma. In the Rainbow Canyon area, such a change is constrained to have occurred after 5.7 Ma, the maximum age of many of the north-northwest–trending dikes.

Wright (1976), from fault slip directions, and Zoback and Zoback (1980), from earthquake focal mechanisms, concluded that the maximum compressive and intermediate stress magnitudes must be about equal in the southwestern Great Basin. Fluctuations in the relative magnitudes of these two stresses could thus account for north-northwest–striking strike-slip faults and nodal planes and north-northeast–striking normal faults and nodal planes. The geologic evidence in Rainbow Canyon supports this idea.

A PALEOMAGNETIC TEST FOR CRUSTAL ROTATION

Because the southwestern Great Basin is a region containing many faults with a demonstrable horizontal component of slip, structural blocks may have been rotated between pairs of faults. Blocks could be rotated ball-bearing style between faults, in which case one might expect clockwise rotations in this right-lateral strike-slip regime (for example, Beck, 1976, 1980; Brown and Golombek, 1985, 1986). Another possibility is that counterclockwise rotation would be required to accommodate regional right-lateral slip between blocks that remain in contact with

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

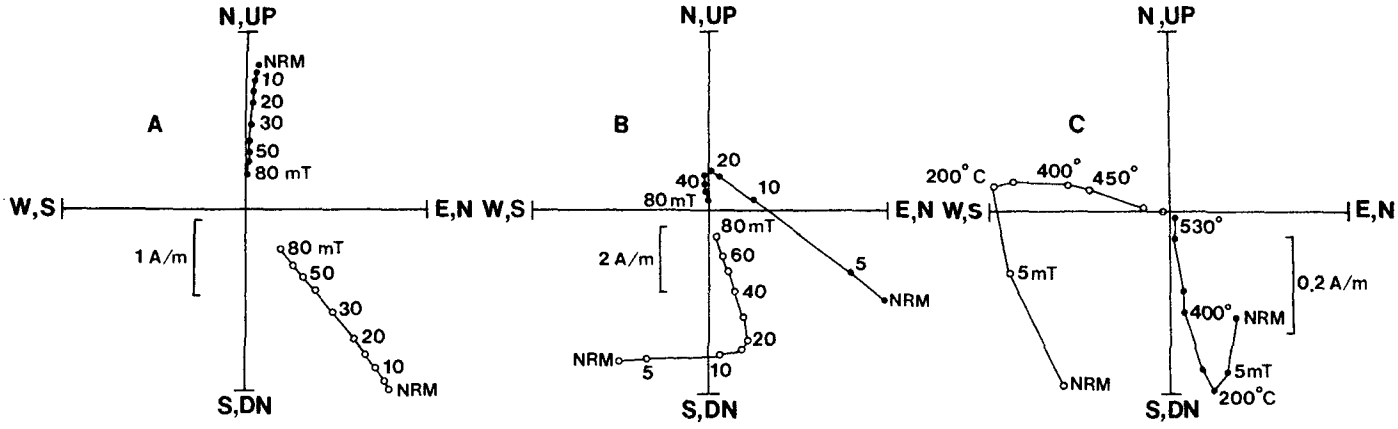


Figure 10. Orthogonal vector plots of demagnetization of basalts in the Rainbow Canyon area. (A) AF (alternating-field) demagnetization of specimen from flow N14, showing one component of magnetization. (B) AF demagnetization of specimen from flow N7, showing two components of magnetization. (C) Thermal demagnetization of specimen from flow R22. A/m, amps/meter; mT, milliTeslas.

some reference boundary (Ron and others, 1984; Garfunkel and Ron, 1985; Nur and others, 1986). The boundary in this case could be the Garlock fault (Fig. 1) and the Mojave block to the south. Either progressive rotation would affect the apparent extension directions as conventionally determined from structural features.

The interpretation of the structural data from Rainbow Canyon is dependent on whether the area has been rotated. To clarify the situation, the rotation hypothesis was tested paleomagnetically. The detailed paleomagnetic examination of the Rainbow Canyon area also tested the validity of the field-determined magnetic polarity sequence in the volcanic section.

Fifty-seven cores were collected from 14 sites in 4 localities. Each site represents a separate flow, which presumably cooled quickly enough that secular variation of the geomagnetic field was not averaged. The only exceptions are 2 sites, separated by about 1 km, that both sample unit N11. A sun compass was used to orient all of the samples, and accuracy of orientation is estimated to be ±2°. The last letter and numbers of site designations in Table 2 correspond to the units identified in Figure 5. The beginning letter or letters refer to the section locations in Figure 4.

Remanence measurements were made on a superconducting magnetometer. Samples were progressively demagnetized using the alternating-field method at peak fields ranging from 5 to 80 milliTeslas (mT).

Characteristic directions were isolated from 12 sites (Table 2). Some display only one component of magnetization (Fig. 10A), but many show evidence of both normal and reversed components of magnetization (Fig. 10B). Samples from the same site consistently show very similar behavior.

Samples from sites WCN11 and HCR22 showed unusually shallow initial inclinations and had not reached a stable end-point direction after demagnetization at 80 mT. Additional samples were therefore progressively thermally demagnetized up to 620 °C. The thermal demagnetization was particularly helpful in removing a low-temperature normal component of magnetization from samples at site HCR22, which had been assigned a normal polarity when measured in the field but was determined to have reversed polarization in the laboratory (Fig. 10C).

No tilt correction was applied to the results because the rocks are assumed to be within 4° of their original orientations.

TABLE 2. RESULTS OF PALEOMAGNETIC STUDY OF RAINBOW CANYON

Site	Polarity	Dec.	Inc.	N	R	k	α_{95}
CR1	Reversed	359.7	52.9	4	3.9954	652	3.6
CTF*	Reversed	359.7	53.8	4	3.9968	938	3.0
CR3	Reversed	358.2	50.8	4	3.9837	184	6.8
CN5	Normal	304.5	66.2	5	4.9971	1379	2.1
CN6	Normal	332.2	62.9	4	3.9867	226	6.1
CN7	Normal	349.9	61.6	4	3.9723	108	8.9
BR8	Reversed	19.1	30.3	4	3.8928	28	17.7
BR9	Reversed	23.8	23.3	4	3.9911	337	5.6
HCR10	Reversed	350.7	50.9	4	3.9924	395	4.6
BN&WCN11	Normal	9.6	54.6	7	6.8975	58	8.0
HCN14	Normal	4.5	51.9	4	3.9981	1548	2.3
HCN20	Normal	352.3	64.3	4	3.9809	157	7.4
HCR22	Reversed	356.1	11.9	5	4.9093	44	11.6
Mean all sites		359.5	50.5	13	12.2483	15	10.7
Mean without site HCR22		0.0	53.6	12	11.4848	20	9.6

Note: N, number of samples; R, resultant vector length; $k = \frac{N-1}{N-R}$, unitless number indicating grouping; α_{95}, radius of circle of 95% confidence.

*CTF is from local thin flow in alluvium between R1 and R2.

Results

Mean magnetic directions and statistics are presented in Table 2 and Figure 11A. Sites WCN11 and BN11, which sampled the same flow, are combined into one site. One sample from WCN11 gave results very different from the others from that site and has been disregarded. The precision parameter, k, is high for

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

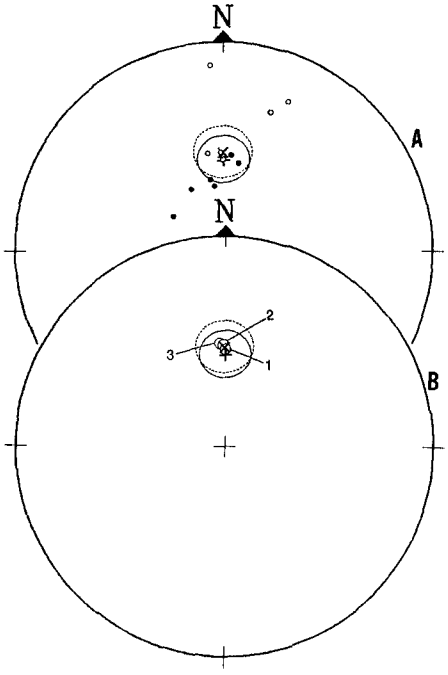


Figure 11. Equal-area plots. X and dashed circles represent, respectively, mean of all sites and 95% confidence interval. + and continuous circle represent mean without site HCR22. (A) Mean magnetic directions of all sites. Solid circles are normal and open circles are reversed directions inverted through the origin. (B) Small circle 1 is expected direction for 0–10 Ma from Brown and Golombek (1985); small circle 2 is for 3–7 Ma from Van Alstine and de Boer (1978); small circle 3 is present-day direction for an axial geocentric dipole field.

most sites, as would be expected for basalts. A high k signifies that the sample directions cluster tightly about the true mean directions (McElhinny, 1973).

Only 13 flows were sampled, and this may be too few to average secular variation. Although some flows may be very close in age to adjacent ones, the total time span represented among the sites is at least 1.8 m.y., long enough to span many cycles of secular variation.

The radii of the circles of 95% confidence, α_{95}, are less than 10° for all sites except BR8 and HCR22. In calculating the mean direction for the entire section, the rejection of both these sites was considered. BR8 has an α_{95} of 17.7°, the largest among all of the sites. The site mean is not as far from the combined mean as that of some of the other sites, however. It also is very close to that of BR9, which had a small α_{95}, and field relations suggest that the rocks from the two sites are very close to each other in age. Site BR8 therefore has been included in all calculations.

Site HCR22 had an α_{95} of 11.6°, the second highest in the study. Its site mean was the farthest from all of the others, and its samples had the least-stable magnetization. The mean direction therefore was calculated in two ways, one using data from all sites and the other excluding site HCR22 (Fig. 11A). The mean direction for all sites has a declination (D) of 359.5° (paleomagnetic declinations being measured clockwise from north) and an inclination (I) of +50.5°. Rejecting site HCR22 results in a mean direction with D = 0.0° and I = +53.6°. These two directions differ by about 3° of arc. The rejection of site HCR22 reduces α_{95} from 10.7° to 9.6°.

Van Alstine and de Boer (1978) determined a paleomagnetic pole for stable North America for the period 3–7 Ma with latitude 88°N and longitude 97°E. This predicts D = 359° and I = 50° at the location of Rainbow Canyon. If the North American pole calculated by Brown and Golombek (1985) for 0–10 Ma (86.5°N, 114.7°E) is used, a predicted direction D = 356.8° and I = 49.3° is calculated. For an axial dipole field, the present direction at Rainbow Canyon should be D = 0.0°, I = +51.7°.

These expected directions are plotted in Figure 11B along with the determined mean directions and their circles of 95% confidence. As can be seen, the predicted directions fall close to the measured site means and within the circles of confidence whether or not site HCR22 is included. If the method of Demarest (1983) and the pole of Brown and Golombek (1985) are used, the rotation, R, is 3.6° and the flattening, F, is −5.8°. The 95% confidence limits are ±13.7° for R and ±10.6° for F. If the extension direction has been west-northwest–east-southeast throughout Neogene time, then the earlier west-southwest–east-northeast extension indicators would be west-northwest–east-southeast indicators rotated counterclockwise. Yet, this study indicates that even at the 95% confidence level, the maximum counterclockwise rotation could be no more than 10.1° (3.6° minus 13.7°). The inferred changes in extension direction thus are likely real rather than apparent. This may also indicate that the slips on regional strike-slip faults or the blocks between faults are small enough that the rotations produced are too small to be resolved. Alternatively, the most obvious boundary, the Mojave block and Garlock fault, is itself being rotated. Additional paleomagnetic data are needed to further constrain these regional tectonic models.

In the flows sampled, the field determinations of magnetic polarities by fluxgate magnetometer are valid except in the case of unit R22, which had a strong normal overprint. R22 gave very shallow inclination values in the field, and its polarity was somewhat suspect at that time.

CONCLUSIONS

The following conclusions are based on the data in this study and on the regional information discussed in earlier sections.

1. Basin-range crustal extension began in the Darwin Plateau area by about 7–8 Ma.
2. The formation of Panamint Valley was underway by 6.1 Ma.
3. A possible switch from west-southwest–east-northeast to west-northwest–east-southeast extension occurred after 5.7 Ma, in agreement with and perhaps constraining ages of such a switch in the central Great Basin.
4. There is evidence that the maximum compressive and intermediate stresses have alternated between vertical and north-northeast in the latest stress regime, in agreement with other types of data in the southwestern Great Basin.
5. Data do not indicate rotation of the Darwin Plateau since 7.7 Ma, although at the 95% confidence level, as much as 10° of counterclockwise or 17° of clockwise rotation is allowed.

ACKNOWLEDGMENTS

The late R. H. Jahns introduced me to the study area. I thank him, G. A. Thompson, G. I. Smith, D. K. Larue, and L. R. Kanter for helpful discussion of this work. S. Barrow, R. Brigham, P. Jansma, Kanter, and D. Merritts provided greatly needed field assistance. Kanter, R. Gordon, D. B. Bieler, G. R. Roquemore, R. H. Hazlett, and an anonymous reviewer critically reviewed this paper.

REFERENCES CITED

Bacon, C. R., Giovannetti, D. M., Duffield, W. A., Dalrymple, G. B., and Drake, R. E., 1982, Age of the Coso Formation, Inyo County, California: U.S. Geological Survey Bulletin 1527, 18 p.

Beck, M. E., Jr., 1976, Discordant paleomagnetic pole positions as evidence of regional shear in the western Cordillera of North America: American Journal of Science, v. 276, p. 694–712.

——1980, Paleomagnetic record of plate-margin tectonic processes along the western edge of North America: Journal of Geophysical Research, v. 85, p. 7115–7131.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Brown, L. L., and Golombek, M. P., 1985, Tectonic rotation within the Rio Grande rift: Evidence from paleomagnetic studies: Journal of Geophysical Research, v. 90, p. 790–802.
——1986, Block rotations in the Rio Grande rift, New Mexico: Tectonics, v. 5, p. 423–438.
Burchfiel, B. C., Hodges, K. V., and Royden, L. H., 1987, Geology of Panamint Valley–Saline Valley pull-apart system, California: Palinspastic evidence for low-angle geometry of a Neogene range-bounding fault: Journal of Geophysical Research, v. 92, p. 10422–10426.
Christiansen, R. L., and Lipman, P. W., 1972, Cenozoic volcanism and plate-tectonic evolution of the western United States. II. Late Cenozoic: Royal Society of London Philosophical Transactions, ser. A, v. 271, p. 249–284.
Demarest, H. H., Jr., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.
Doell, R. R., and Cox, A., 1967, Measurement of natural remanent magnetization at the outcrop, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, the Netherlands, Elsevier, p. 159–162.
Eaton, G. P., 1982, The Basin and Range province: Origin and tectonic significance: Annual Review of Earth and Planetary Sciences, v. 10, p. 409–440.
Garfunkel, Z., and Ron, H., 1985, Block rotation and deformation by strike-slip faults, 2. The properties of a type of macroscopic discontinuous deformation: Journal of Geophysical Research, v. 90, p. 8589–8602.
Hall, W. E., 1971, Geology of the Panamint Butte quadrangle, Inyo County, California: U.S. Geological Survey Bulletin 1299, 67 p.
Hall, W. E., and MacKevett, E. M., Jr., 1962, Geology and ore deposits of the Darwin quadrangle, Inyo County, California: U.S. Geological Survey Professional Paper 368, 87 p.
Harland, W. B., Cox, A. V., Llewellyn, P. G., Pickton, C.A.G., Smith, A. G., and Walters, R., 1982, A geologic time scale: Cambridge, United Kingdom, Cambridge University Press, 131 p.
Hopper, R. H., 1947, Geologic section from the Sierra Nevada to Death Valley, California: Geological Society of America Bulletin, v. 58, p. 393–432.
Irvine, T. N., and Baragar, W.R.A., 1971, A guide to the chemical classification of the common volcanic rocks: Canadian Journal of Earth Sciences, v. 8, p. 523–548.
Larsen, N. W., 1979, Chronology of late Cenozoic basaltic volcanism along a segment of the Sierra Nevada and Basin and Range province boundary [Ph.D. thesis]: Provo, Utah, Brigham Young University, 95 p.
Lubetkin, L., and Clark, M. M., 1985, Late Quaternary activity along the Lone Pine fault, eastern California, *in* Stein, R. S., and Bucknam, R. C., eds., Proceedings, Workshop XXVIII, On the Borah Peak, Idaho, earthquake (Volume A): U.S. Geological Survey Open-File Report 85-290, p. 118–140.
McElhinny, M. W., 1973, Paleomagnetism and plate tectonics: Cambridge, United Kingdom, Cambridge University Press, 358 p.
Michael, A. J., 1984, Determination of stress from slip data: Faults and folds: Journal of Geophysical Research, v. 89, p. 11517–11526.
Massachusetts Institute of Technology 1985 Field Geophysics Course and Biehler, S., 1987, A geophysical investigation of the northern Panamint Valley, Inyo County, California: Evidence for possible low-angle normal faulting at shallow depth in the crust: Journal of Geophysical Research, v. 92, p. 10427–10441.
Nur, A., Ron, H., and Scotti, O., 1986, Fault mechanics and the kinematics of block rotations: Geology, v. 14, p. 746–749.
Ron, H., Freund, R., Garfunkel, Z., and Nur, A., 1984, Block rotation by strike-slip faulting: Structural and paleomagnetic evidence: Journal of Geophysical Research, v. 89, p. 6256–6270.
Roquemore, G. R., 1980, Structure, tectonics, and stress field of the Coso Range, Inyo County, California: Journal of Geophysical Research, v. 85, p. 2434–2440.
Schweig, E. S., 1982, Late Cenozoic stratigraphy and tectonics of the Darwin Plateau, Inyo County, California [M.S. thesis]: Stanford, California, Stanford University, 85 p.
——1986, The inception of Basin and Range tectonics in the region between Death Valley and the Sierra Nevada, California: Geological Society of America Abstracts with Programs, v. 18, p. 181–182.
Smith, R.S.U., 1975, Late Quaternary pluvial and tectonic history of Panamint Valley, Inyo and San Bernardino Counties, California [Ph.D. thesis]: Pasadena, California, California Institute of Technology, 301 p.
Steiger, R. H., and Jäger, E., 1977, Subcommission on Geochronology: Convention on the use of decay constants in geo- and cosmochronology: Earth and Planetary Science Letters, v. 36, p. 359–362.
Stewart, J. H., 1978, Basin-range structure in western North America: A review, *in* Smith, R. B., and Eaton, G. P., eds., Cenozoic tectonics and regional geophysics of the western Cordillera: Geological Society of America Memoir 152, p. 1–31.
Van Alstine, D. R., and de Boer, J., 1978, A new technique for constructing apparent polar wander paths and the revised Phanerozoic path for North America: Geology, v. 6, p. 137–139.
Wright, L. A., 1976, Late Cenozoic fault patterns and stress fields in the Great Basin and westward displacement of the Sierra Nevada block: Geology, v. 4, p. 489–494.
Wright, L. A., Drake, R. E., and Troxel, B. W., 1984, Evidence for the westward migration of severe Cenozoic extension, southwestern Great Basin, California: Geological Society of America Abstracts with Programs, v. 16, p. 701.
Zellmer, J. T., 1980, Recent deformation in the Saline Valley region, Inyo County, California [Ph.D. thesis]: Reno, Nevada, University of Nevada, 168 p.
Zoback, M. L., and Thompson, G. A., 1978, Basin and Range rifting in northern Nevada: Clues from a mid-Miocene rift and its subsequent offset: Geology, v. 6, p. 111–116.
Zoback, M. L., and Zoback, M. D., 1980, State of stress in the conterminous United States: Journal of Geophysical Research, v. 85, p. 6113–6156.
——1989, Tectonic stress field of the continental U.S., *in* Pakiser, L., and Mooney, W., eds., Geophysical framework of the continental United States: Geological Society of America Memoir (in press).
Zoback, M. L., Anderson, R. E., and Thompson, G. A., 1981, Cainozoic evolution of the state of stress and style of tectonism of the Basin and Range province of the western United States: Royal Society of London Philosophical Transactions, ser. A, v. 300, p. 407–434.

Manuscript Received by the Society May 11, 1988
Revised Manuscript Received September 30, 1988
Manuscript Accepted October 3, 1988

Printed in U.S.A.